

LAB SHEET



K.R. MANGALAM UNIVERSITY

SOET

(School of Engineering & Technology)

B. TECH CSE (CORE)

Database & Management System Lab (ENCS254)

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LAB SHEET – 4

Title: Implementation of **User Creation, Deletion, and Privilege Management in SQL**

Problem Statement: Design and execute SQL commands to **create users, manage their access privileges, and remove users** in a database system. This assignment focuses on **database security and access control mechanisms**.

Database security ensures that only authorized users can access or modify data. MySQL provides commands such as:

- CREATE USER → to create new users
- GRANT → to give permissions
- REVOKE → to remove permissions
- DROP USER → to delete users

Privileges include:

- SELECT (read data)
- INSERT (add data)
- UPDATE (modify data)
- DELETE (remove data)

Step 1: Create Database and Tables

- `CREATE DATABASE LAB_SHEET_4;`
- `USE LAB_SHEET_4;`

Explanation:

- Creates a database named LAB_SHEET_4
- Selects it for further operations

Create Tables

- ```
CREATE TABLE Student (
 id INT PRIMARY KEY,
 name VARCHAR(50),
 age INT
);
```

**Explanation:**

Creates a Student table where:

- id is unique
- name stores student name
- age stores age

```
• CREATE TABLE Course (
 course_id INT PRIMARY KEY,
 course_name VARCHAR(50)
);
```

**Explanation:**

Creates a Course table to store course details.

## Part A: User Creation

**Step 2: Create user1**

```
• CREATE USER 'user1'@'localhost' IDENTIFIED BY 'pass123';
```

**Explanation:**

- Creates a user named user1
- 'localhost' means user can log in from same system
- IDENTIFIED BY sets password

**Step 3: Create user2**

```
• CREATE USER 'user2'@'localhost' IDENTIFIED BY 'pass124';
```

**Explanation:**

Creates another user user2 with login credentials.

## Part B: Granting Privileges

**Step 4: Grant SELECT and INSERT to user1**

```
• GRANT SELECT, INSERT ON LAB_SHEET_4.Student TO 'user1'@'localhost';
```

### Explanation:

- SELECT → allows viewing data
- INSERT → allows adding data
- Access is limited to Student table only

### Step 5: Grant ALL privileges to user2

- `GRANT ALL PRIVILEGES ON LAB_SHEET_4.Course TO 'user2'@'localhost';`

### Explanation:

- Gives full access (SELECT, INSERT, UPDATE, DELETE, etc.)
- Only on Course table

### Step 6: WITH GRANT OPTION

- `GRANT SELECT, INSERT ON LAB_SHEET_4.Student TO 'user1'@'localhost' WITH GRANT OPTION;`

### Explanation:

- Allows user1 to **give permissions to other users**
- This is called **privilege delegation**

## Part C: Revoking Privileges

### Step 7: Revoke INSERT from user1

- `REVOKE INSERT ON LAB_SHEET_4.Student FROM 'user1'@'localhost';`

### Explanation:

- Removes only INSERT permission
- user1 can now only read data

### Step 8: Revoke ALL from user2

- `REVOKE ALL PRIVILEGES ON LAB_SHEET_4.Course FROM 'user2'@'localhost';`

### Explanation:

- Removes all permissions
- user2 cannot access Course table anymore

## Part D: User Deletion

### Step 9: Drop user2

- `DROP USER 'user2'@'localhost';`

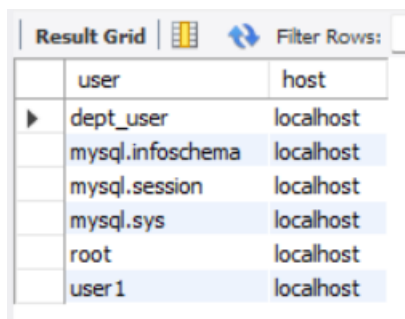
#### Explanation:

- Permanently deletes the user
- All privileges are removed

### Step 10: Show all users

- `SELECT user, host FROM mysql.user;`

#### Output:



|   | user             | host      |
|---|------------------|-----------|
| ▶ | dept_user        | localhost |
|   | mysql.infoschema | localhost |
|   | mysql.session    | localhost |
|   | mysql.sys        | localhost |
|   | root             | localhost |
|   | user1            | localhost |

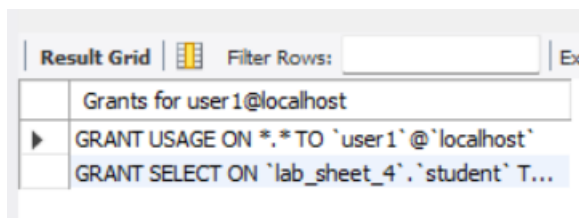
#### Explanation:

Displays all users present in MySQL system.

### Step 11: Show privileges of user1

- `SHOW GRANTS FOR 'user1'@'localhost';`

#### Output:



|   | Grants for user1@localhost                   |
|---|----------------------------------------------|
| ▶ | GRANT USAGE ON *.* TO 'user1'@'localhost'    |
|   | GRANT SELECT ON `lab_sheet_4`.`student` T... |

#### Explanation:

Displays all permissions assigned to user1.